



FACULTY OF
**COMPUTER
SCIENCE**

Tutorial Kuesioner dan Tool PLS-SEM

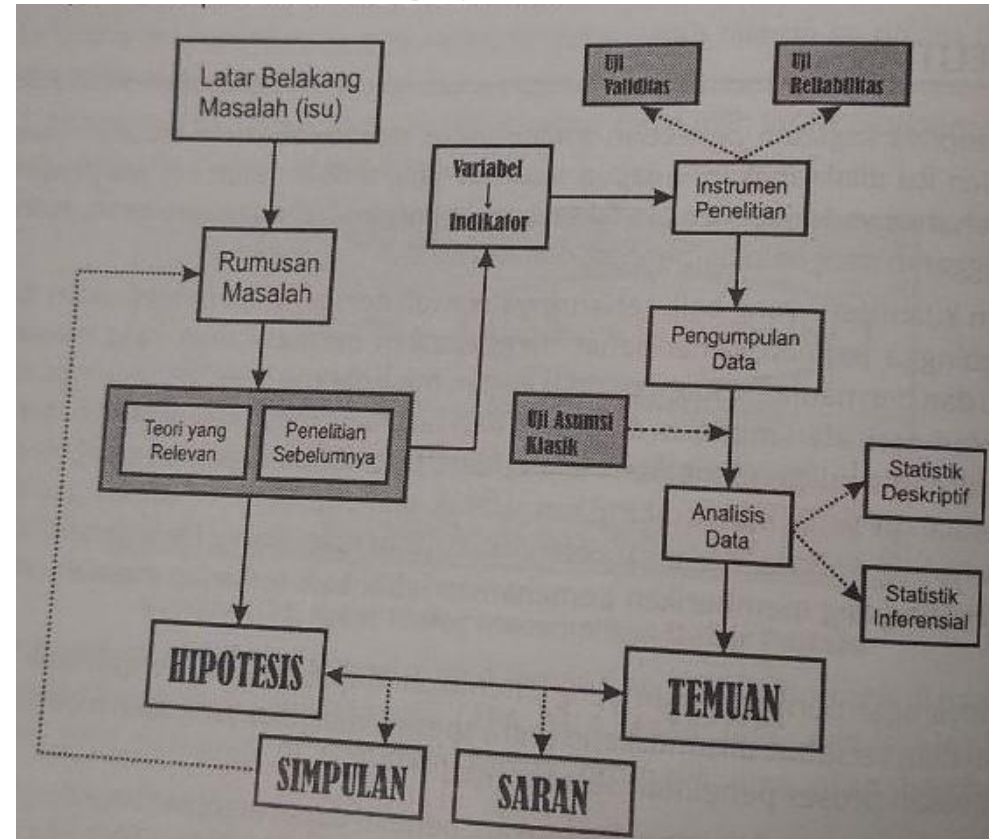
Damayanti Elisabeth, S.Komp., M. Kom.

Manajemen Pengetahun

Magister Teknologi Informasi

2024-2025

Research Methodology



Proses Penelitian

Wiyono, G. (2020). Merancang penelitian bisnis dengan alat analisis SPSS 25 & SmartPLS 3.2. 8.

Research Methodology

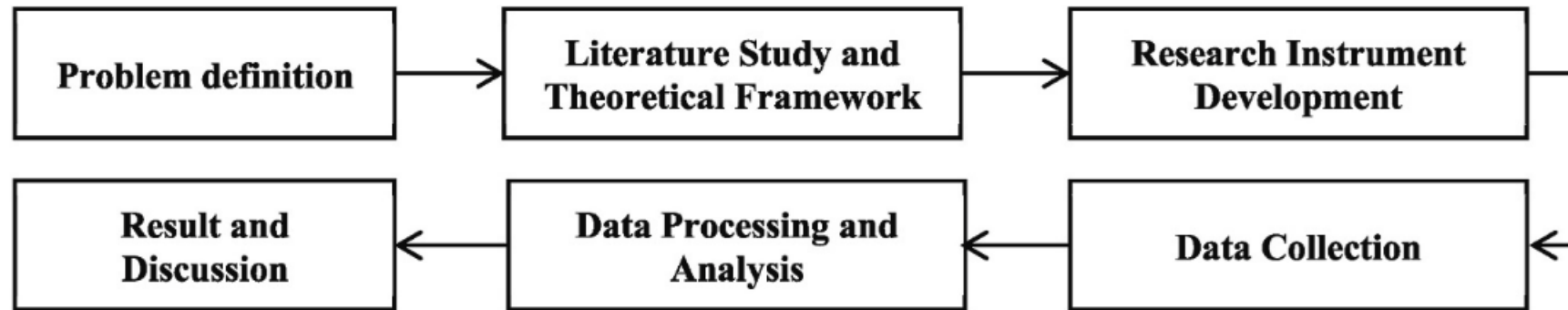


Fig. 2. Research methodology.

Fitriah, N., Budi, A., Adnan, H. R., Firmansyah, F., Hidayanto, A. N., Kurnia, S., & Purwandari, B. (2021).

Technology in Society Why do people want to use location-based application for emergency situations ?

The extension of UTAUT perspectives. *Technology in Society*, 65(February), 101480.

<https://doi.org/10.1016/j.techsoc.2020.101480>

Multivariate data analysis – Research Purpose Examples

- Joseph, F. H. J. R., Barry, J. B., Rolph, E. A., & Rolph, E. A. (2010). *Multivariate data analysis*. Pearson Prentice Hall.
- The purpose of this study is to determine the factors that influence user intention to use the X-Igent Panic Button application in an emergency
 - Fitriah, N., Budi, A., Adnan, H. R., Firmansyah, F., Hidayanto, A. N., Kurnia, S., & Purwandari, B. (2021). **Technology in Society Why do people want to use location-based application for emergency situations ? The extension of UTAUT perspectives.** *Technology in Society*, 65(February), 101480. <https://doi.org/10.1016/j.techsoc.2020.101480>
- This study investigated the different of the significant antecedent factor which is influenced the costumer attitude toward advertising in each of the three popular media (SMS, e-mail and social media) with case study in Indonesia.
 - Noprisson, H., Husin, N., Zulkarnaim, N., Rahayu, P., Ramadhan, A., & Sensuse, D. I. (2017). **Antecedent factors of consumer attitudes toward SMS, E-mail and social media for advertising.** *2016 International Conference on Advanced Computer Science and Information Systems, ICACSI 2016*, 165–170. <https://doi.org/10.1109/ICACSI.2016.7872742>

Multivariate data analysis examples

Fitriah, N., Budi, A., Adnan, H. R., Firmansyah, F.,
 Hidayanto, A. N., Kurnia, S., & Purwandari, B. (2021).
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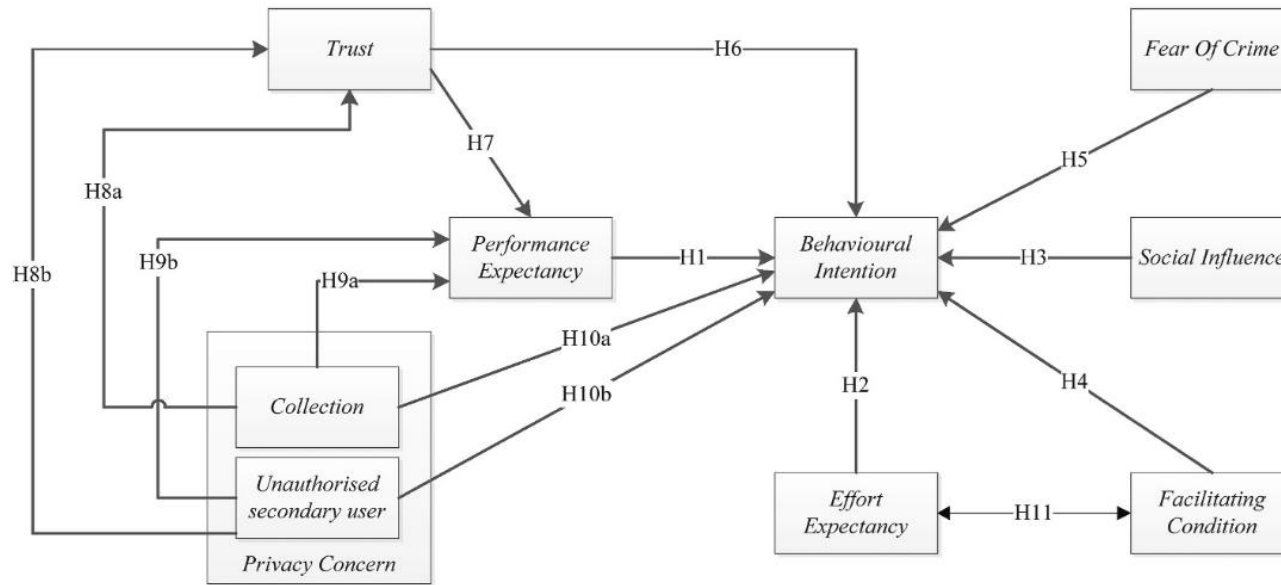


Fig. 1. Research model.

Noprisson, H., Husin, N., Zulkarnaim, N., Rahayu, P.,
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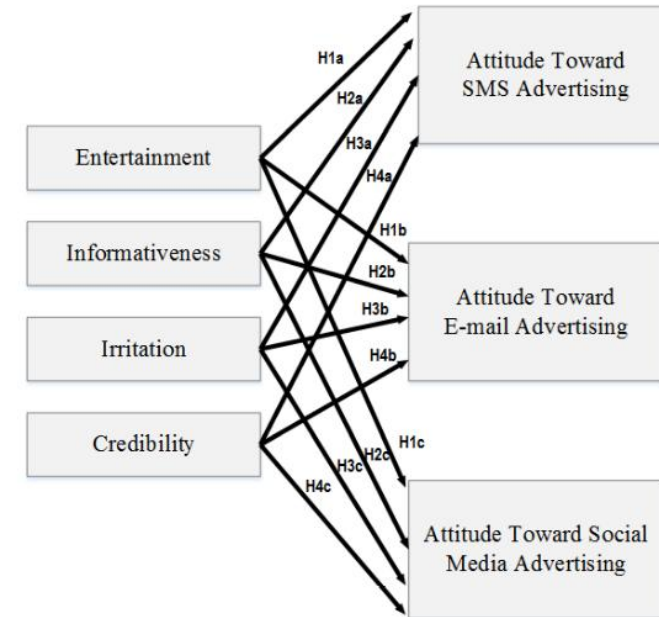


Fig. 3. Model for investigating antecedent factors of consumer attitudes towards (SMS/E-mail/Social Media) advertising

Research Instrument Development

Example: UTAUT

Venkatesh, V., Thong, J. Y., & Xu, X. (2012). Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS quarterly*, 157-178.

Venkatesh et al./Consumer Acceptance and Use of IT

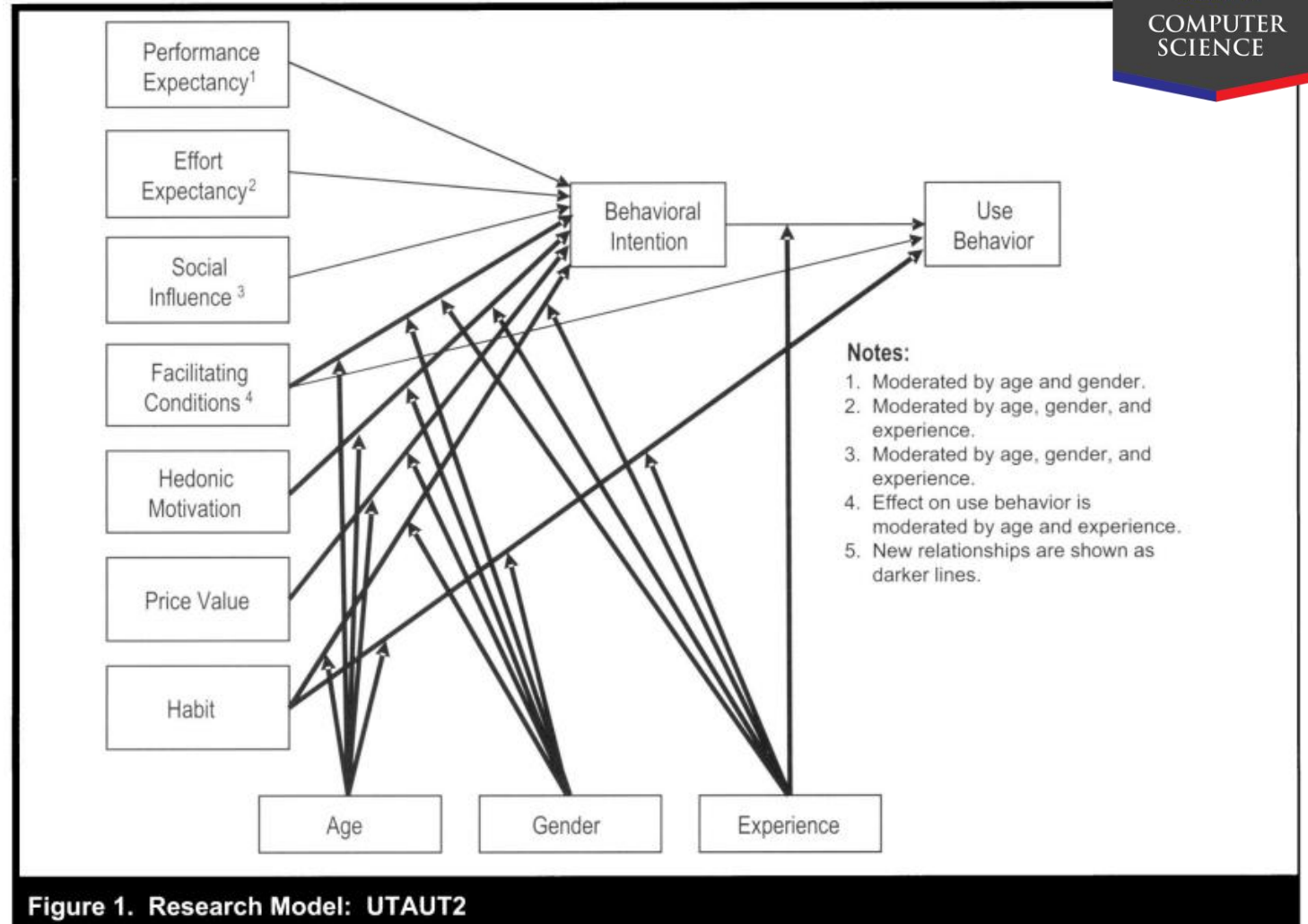


Figure 1. Research Model: UTAUT2

Research Instrument Development (2)

Fitriah, N., Budi, A., Adnan, H. R., Firmansyah, F., Hidayanto, A. N., Kurnia, S., & Purwandari, B. (2021). Technology in Society Why do people want to use location-based application for emergency situations ? The extension of UTAUT perspectives. *Technology in Society, 65*(February), 101480. <https://doi.org/10.1016/j.techsoc.2020.101480>

Table 2
Summary of previous studies.

No	Author, Year	Country	Study Context	Employed Theories	Methodology	Model/List of Factor Proposed	Key Findings	Limitation/Future Direction
1	(Aloudat, Michael, Chen, & Al-Debei, 2014)	Australia	Explore general factors that influence the acceptance of government mobile LBS application	TAM and Additional Factors	Quantitative; Survey research; PLS-SEM Data Analysis	1) Perceived Usefulness 2) Perceived Ease of Use 3) Perceived Risk 4) Trust 5) Privacy Concern (Collection, unauthorized secondary use) 6) Visibility 7) Perceived Service Quality (Responsiveness, currency, accuracy)	These factors: 1) Perceived Usefulness; 2) Trust; 3) Privacy Concern (Collection, unauthorized secondary use), influence LBS adoption	1) Use more combination of survey techniques 2) Explore context outside Australia by considering government or cultural aspects
2	(Fodor & Brem, 2015)	Germany	Compare two models of privacy concerns on LBS user adoption	IUIPC Framework, CFIP Framework	Quantitative; Survey research; Harman's single-factor test; exploratory factor analysis (EFA); SEM PLS	1. CFIP (collection, control, and awareness) 2. IUIPC (collection, errors, improper access, and secondary use) 3. Perceived Risk	1. Collection influence trust 2. Error influence perceived risk 3. Collection influence perceived risk 4. Trust influence perceived risk	1) Exploration in other geographic 2) Need to explore the actual behavior, not just the intention

Research Instrument Development (2)

Fitriah, N., Budi, A., Adnan, H. R., Firmansyah, F., Hidayanto, A. N., Kurnia, S., & Purwandari, B. (2021). Technology in Society Why do people want to use location-based application for emergency situations ? The extension of UTAUT perspectives. *Technology in Society*, 65(February), 101480. <https://doi.org/10.1016/j.techsoc.2020.101480>

The purpose of this study is to determine the factors that influence user intention to use the X-Igent Panic Button application in an emergency

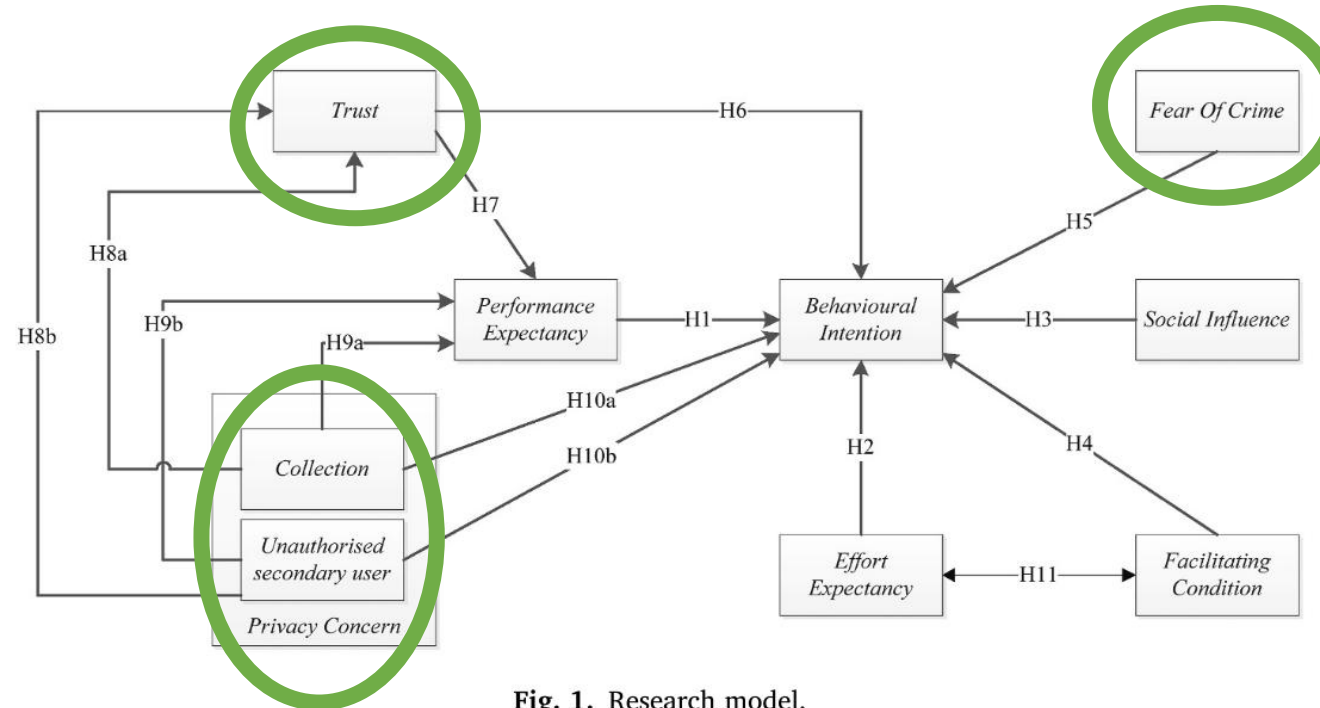
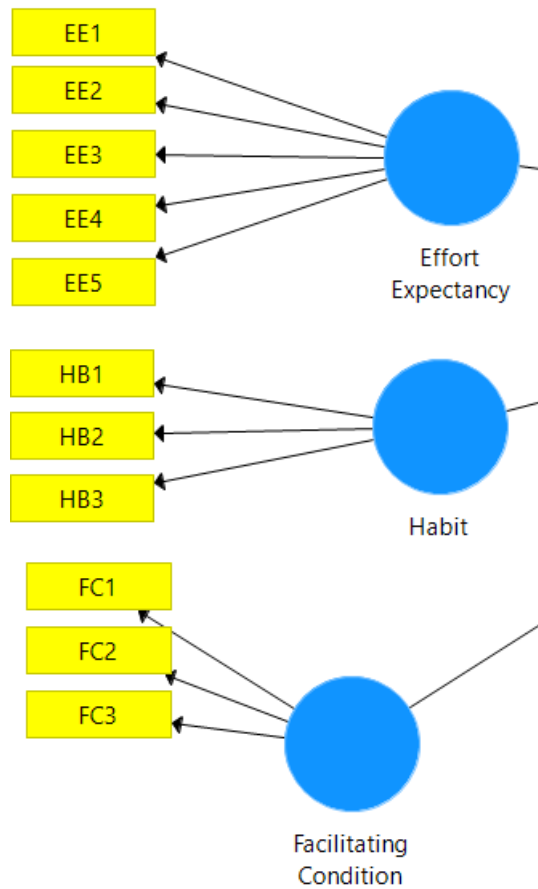


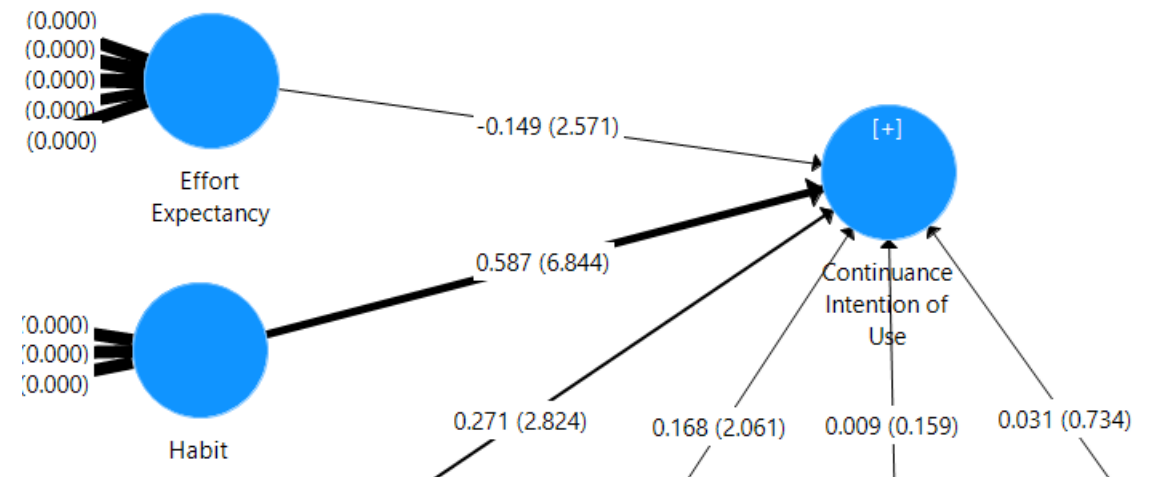
Fig. 1. Research model.

Pengukuran Indikator

Pengukuran Reflektif

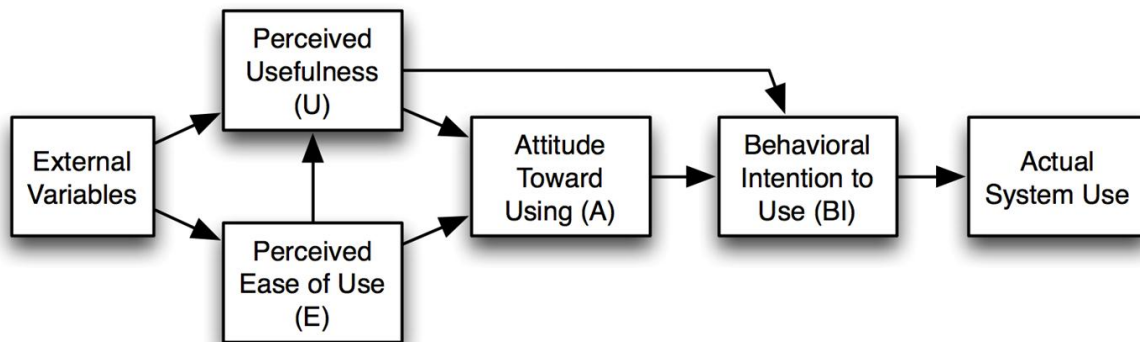


Pengukuran Formatif



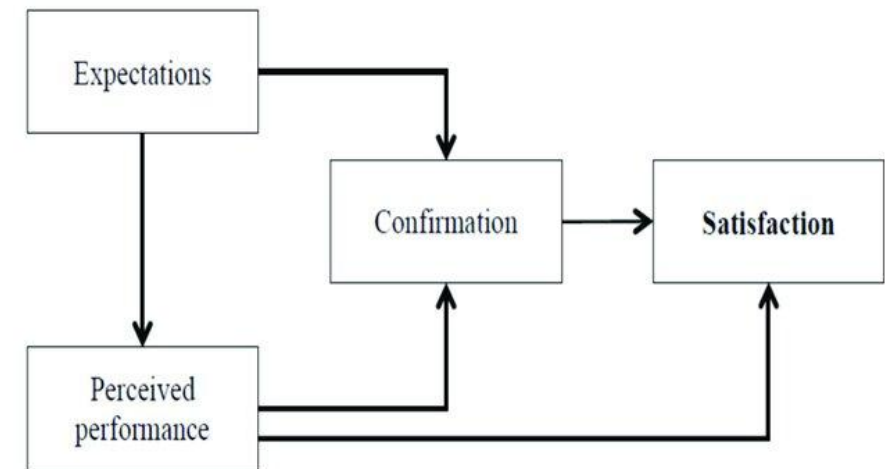
Multivariate data analysis (*don't!*)

- Terdapat 2 pengujian penelitian dalam 1 kuesioner
- Contoh menguji 2 produk dalam 1 model
- Variabel serupa atau saling berkaitan digunakan (contoh: Tahun Lahir & Umur)
- User memikirkan dan mengisi banyak pertanyaan (contoh: 40 pertanyaan)



Model Penerimaan teknologi

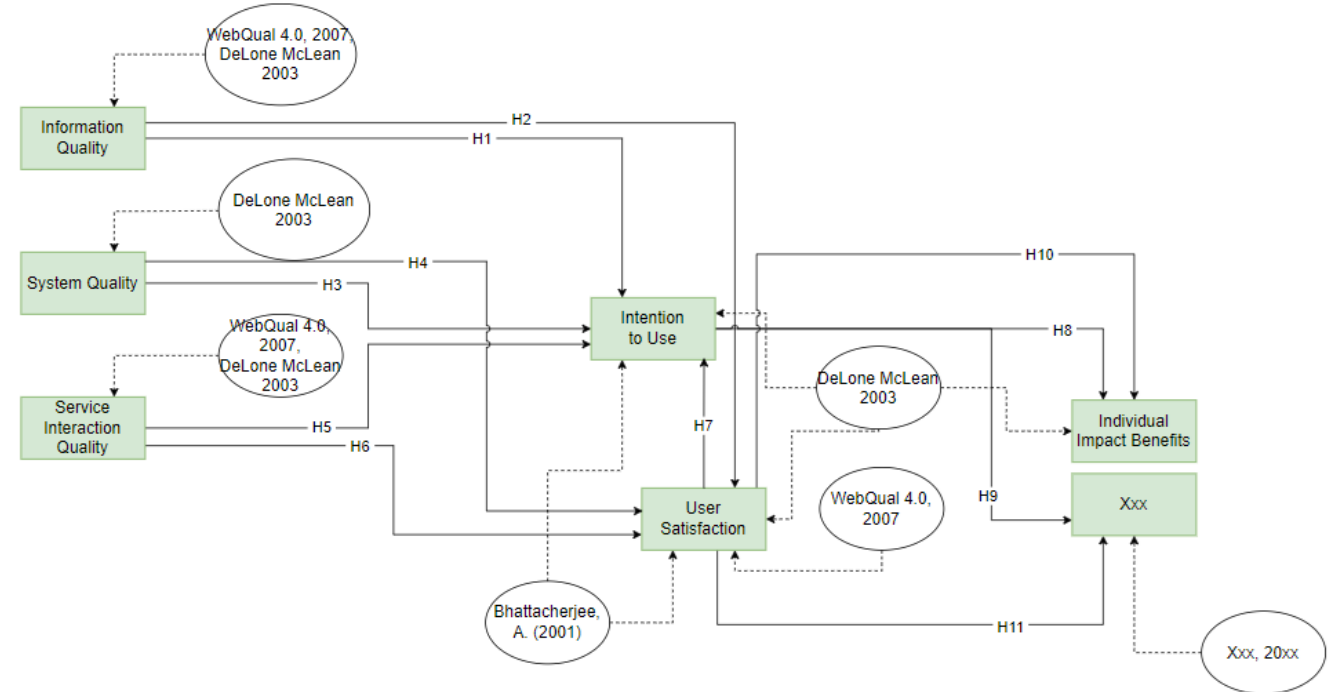
&



Model Konfirmasi Ekspektasi

Multivariate data analysis (*do!*)

- Buat Model dari teori yang sudah ada
- Bangun pernyataan kuesioner dari definisi dan juga target penelitian
- Perhatikan isi data variabel
- Target penelitian 1 saja
- Model juga 1 saja



Model Kualitas Sistem Informasi

Contoh Hipotesis Kualitas Sistem Informasi

H1: "**Kualitas informasi** dari Fitur X memiliki pengaruh positif yang signifikan terhadap **niat pengguna** untuk menggunakan Fitur X."

H2: "**Kualitas informasi** dari Fitur X memiliki dampak positif yang signifikan terhadap **kepuasan pengguna** terhadap Fitur X."

Contoh Pernyataan Survei UTAUT

- **Performance Expectancy**

- PE1. I find the mobile Internet **useful** in my daily life.
- PE2. Using the mobile Internet **increases my chances of achieving things** that are important to me. (dropped)
- PE3. Using the mobile Internet helps me accomplish things more quickly.
- PE4. Using the mobile Internet increases my productivity.

- **Effort Expectancy**

- EE1 . Learning how to use the mobile Internet is easy for me.
- EE2. My interaction with the mobile Internet is clear and understandable.
- EE3. I find mobile Internet easy to use.
- EE4. It is easy for me to become skillful at using the mobile Internet.

- **Social Influence**

- SI1. People who are important to me think that I should use mobile Internet.
- SI2. People who influence my behavior think that I should use mobile Internet.
- SI3. People whose opinions that I value prefer that I use mobile Internet.

- **Facilitating Conditions**

- FC1 . I have the resources necessary to use the mobile Internet.
- FC2. I have the knowledge necessary to use the mobile Internet.
- FC3. Mobile Internet is compatible with other technologies I use.
- FC4. I can get help from others when I have difficulties using the mobile Internet.

- **Hedonic Motivation**

- HM1 . Using the mobile Internet is fun.
- HM2. Using the mobile Internet is enjoyable.
- HM3. Using the mobile Internet is very entertaining.

- **Price Value**

- PV1. Mobile Internet is reasonably priced.
- PV2. Mobile Internet is a good value for the money.
- PV3. At the current price, mobile Internet provides a good value.

- **Habit**

- HT1. The use of mobile Internet has become a habit for me.
- HT2. I am addicted to using the mobile Internet.
- HT3. I must use mobile Internet.
- HT4. Using the mobile Internet has become natural to me. (dropped)

- **Behavioral Intention**

- BI1. I intend to continue using the mobile Internet in the future.
- BI2. I will always try to use the mobile Internet in my daily life.
- BI3. I plan to continue to use the mobile Internet frequently.

Venkatesh, V., Thong, J. Y., & Xu, X. (2012). Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS quarterly*, 157-178.

Contoh Uraian Variabel

Best Practices: Dokumen rujukan untuk pembuatan Research Instrument yang berkualitas adalah menyampaikan rincian variable dan kalimat pernyataan survei

Nguyen, T. T., Phan, D. M., Le, A. H., & Nguyen, L. T. N. (2020). The determinants of citizens' satisfaction of E-government: an empirical study in Vietnam. *The Journal of Asian Finance, Economics and Business*, 7(8), 519-531.

Table 1: Variables Summary

Variable	Meaning	Determined by	Model	Role
SAT	Satisfaction	Overall satisfaction	SAT1	Dependent variable
EFF	Efficiency	Web interface	EFF1	Independent variable
		Search Engine	EFF2	
		Customization	EFF3	
		Appropriate detailed information	EFF4	
		Fresh information	EFF5	
		Ease of use	EFF6	
TRU	Trust	Secure data	TRU1	Independent variable
		Account privacy	TRU2	
		Information privacy	TRU3	
REL	Reliability	Accessibility	REL1	Independent variable
		On-time delivery	REL2	
		Quick forms download	REL3	
		Browser-system compatibility	REL4	
		Freeze site	REL5	
SUP	Citizen Support	Honesty	SUP1	Independent variable
		Quick Support	SUP2	
		Knowledge	SUP3	
		Inspires trustworthiness	SUP4	
CON	Convenience	Access anytime	CON1	Independent variable
		Access anywhere	CON2	
		Convenience to use	CON3	
TRA	Transparency	Reduce bribes	TRA1	Independent variable
		Reduce cost	TRA2	
		Reduce bureaucratic process	TRA3	
		Improve transparency	TRA4	
GEN	Gender	Male	GEN	Control variable
		Female		
AGE	Age	Under 30	AGE	Control variable
		30 to 50		
		Over 50		
EDU	Education	Highschool or lower	EDU	Control variable
		Undergraduate		
		Postgraduate or higher		
INT	Internet Frequency	Less than once a week	INT	Control variable
		2 to 7 times a week		
		More than 7 times a week		



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Pembuatan Survei

- Kalimat Konsisten (objek + positif) -> mempengaruhi arti 2 tail, kemudian hasil validity
- Perhatikan S P O K
- Hindari plural sentences dalam 1 pernyataan
- Tiap pernyataan mewakili 1 indikator
- Tiap Faktor, best practice-nya 3-4 indikator
- Perhatikan kebosanan pengisi

Pemilihan Like-rt

- Skala Likert adalah salah satu alat pengukuran psikologis, kecenderungan, pendapat, subjektif.
- Skala 5 biasa digunakan

Strongly Disagree 1	Disagree 2	Neither Agree Nor Disagree 3	Agree 4	Strongly agree 5
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- Skala 7 lebih sensitif
 - (1) sangat tidak setuju ,(2) tidak setuju ,(3) cukup tidak setuju,(4) Netral,(5) cukup setuju,(6) setuju,(7) Sangat setuju

Tips lainnya membuat kuesioner

Best practice for organising the whole questionnaire

This guidance has been summarised from Gehlbach and Artino (2018).

[Collapse all](#)

Ask the more important items earlier in the questionnaire -

This will increase the likelihood that respondents will answer these questions whilst they are focused and have energy.

Ensure each item applies to each respondent -

Make sure that item content applies to each respondent and is worded accessibly, or you risk alienating respondents to whom the item does not apply (Dillman, Smyth, & Christian, 2014).

Use scales rather than single items -

Scales bolster accuracy, with each question addressing a representative cross-section of the experience.

Maintain a consistent visual layout of the questionnaire -

This helps maintain clarity for the respondent, who can then complete the questionnaire more efficiently.

Place sensitive items (e.g. demographic questions) later in the questionnaire -

Respondents will feel more comfortable sharing this information later on in the questionnaire.

<https://www.imperial.ac.uk/education-research/evaluation/tools-and-resources-for-evaluation/questionnaires/best-practice-in-questionnaire-design/>

Penentuan Sampel

- Tentukan dulu purposive target (contoh: departemen, kota)
- Tentukan jumlah sampel dengan (default):
 - Confidence Level (%) 95
 - Margin of Error (%) 5
- Jika tidak ada populasi harap pilih 200 pengisi kuesioner yang valid

Tools:

[Sample Size Calculator](https://www.calculator.net/sample-size-calculator.html) (<https://www.calculator.net/sample-size-calculator.html>)

Tips lain utk sampling: <https://www.surveymonkey.com/mp/sample-size/>

Penentuan Sampel (Contoh)

Kota Administrasi	Jumlah Populasi 2020	Persentase dari Populasi	Sampel
Jakarta Barat	2,434,511	23%	89
Jakarta Pusat	1,056,896	10%	39
Jakarta Selatan	2,226,812	21%	81
Jakarta Timur	3,037,139	29%	111
Jakarta Utara	1,778,981	17%	65
Kepulauan Seribu	27,749	0%	1
Jumlah Populasi DKI Jakarta	10,562,088		
Confidence Level (%)	95		
Margin of Error (%)	5		
Sample Size	385		385

Demografi

Table 2
Sample profile.

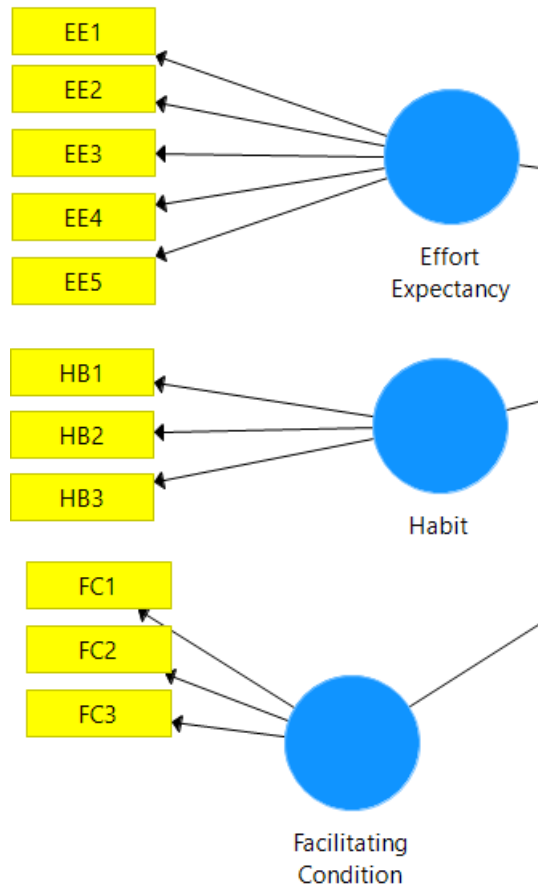
		Frequency	Percentage (%)			Frequency	Percentage (%)
City	Shenyang	262	15.9	Occupation	Retired	50	3
	Changsha	278	16.8		Student	98	6
	Chengdu	302	18.3		Other	32	2
	Shanghai	384	23.2				
Household	Shenzhen	424	25.8	Education	Middle school	110	6.7
	Native	992	60.1		High school	205	12.4
	Non-Native	658	39.9		Bachelor	1106	67
Gender	Male	885	53.6		Postgraduate	229	13.8
	Female	855	46.4	Income (yuan/month)	= < 1000	222	13.5
Age	= < 20	35	2.1		1,001–3,000	312	18.9
	20–29	706	42.8		3,001–5,000	506	30.6
	30–39	602	36.4		> = 5,001	610	37
	40–49	201	12.2	Daily Internet Usage	= < 2 hours	231	14
	50–59	71	4.3		2–4 hours	590	35.8
	> = 60	35	2.2		5–7 hours	481	29.2
Occupation	Government	118	7.1		> = 8 hours	348	21
	Shiye Unit	182	11	Internet Experience	= < 1 year	30	1.8
	Private Company	546	33.1		1–5 years	261	15.8
	Commercial and Service	309	18.7		6–10 years	612	37.1
	Self-Employed	285	17.3		> = 10 years	747	45.3
	Unemployed	30	1.8				

Best Practices: Variabel **City** jika digunakan untuk purposive sampling atau stratified random sampling, statistik harus memenuhi minimal hasil perhitungan sampling

Li, Y., & Shang, H. (2020). Service quality, perceived value, and citizens' continuous-use intention regarding e-government: Empirical evidence from China. *Information & Management*, 57(3), 103197.

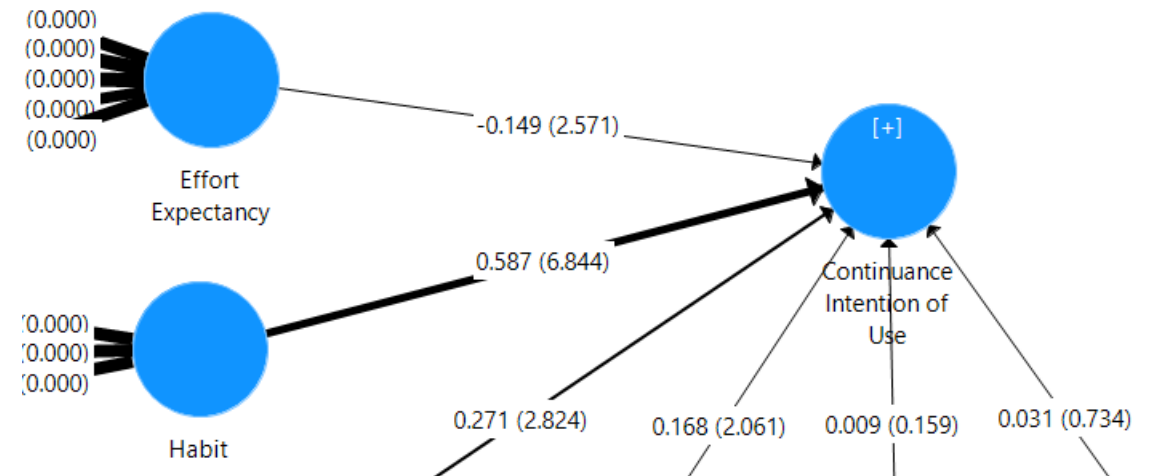
SEM Validity = Measurement Validity + Structural Validity

Measurement Validity

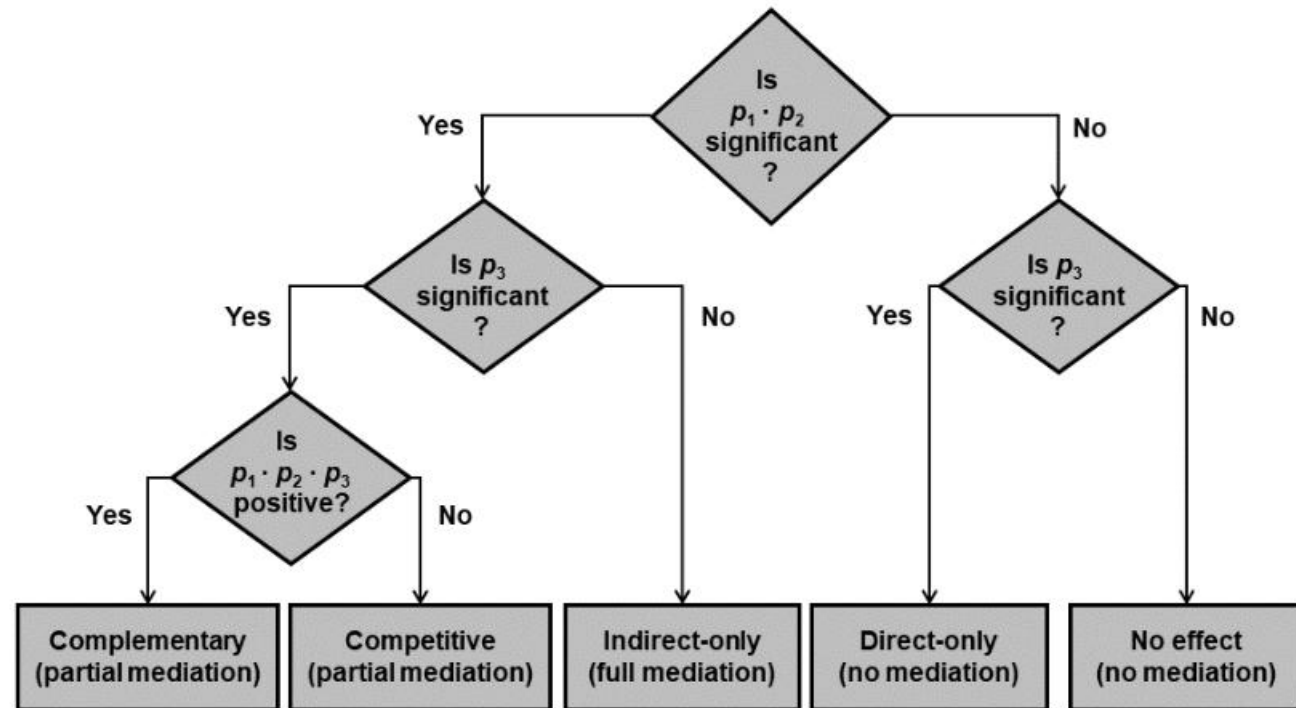
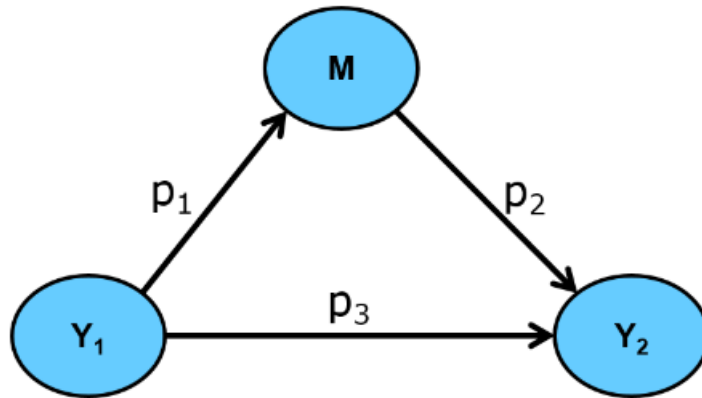


Structural Validity

One Tail vs Two Tail



Mediation in PLS-SEM



Bahan Kualitatif

Pembuatan Rekomendasi Strategi

- **Behavioral Intention**

- BI1. I intend to continue using the mobile Internet in the future.
- BI2. I will always try to use the mobile Internet in my daily life.
- BI3. I plan to continue to use the mobile Internet frequently.

Best Practices: Hasil akhir “faktor/fitur signifikan”, kemudian diterjemahkan menjadi strategi/fitur Knowledge Management Strategies/System

Di masa depan, saya bermaksud untuk **terus menggunakan** informasi pengaduan dari pemerintah

Sehari-hari, saya akan **selalu mencoba** menggunakan informasi pengaduan dari pemerintah

Saya **berencana untuk terus** menggunakan informasi pengaduan dari pemerintah sebagai pedoman untuk menyampaikan aduan

Informasi pengaduan yang disediakan pemerintah **mudah diakses**

Informasi pengaduan pemerintah memudahkan saya **memahami** fasilitas pengaduan pemerintah

Informasi pengaduan pemerintah memudahkan saya **menggunakan** fasilitas pengaduan pemerintah

Sarana informasi pengaduan dari pemerintah **responsif atau cepat tanggap**

Saya **mudah terampil** menggunakan fasilitas pengaduan berkat informasi yang diberikan oleh pemerintah

